Enhancing Rotten Fruit Detection Using Deep Learning and Gaussian Filtering

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***Abstract*— Fruit spoilage in post-harvest handling is a common issue in the agricultural sector, resulting in economic losses and reduced food quality. Manual inspection is a traditional method for detecting rotten fruit but is inefficient for industrial applications. In this paper, a method for detecting rotten fruit using deep learning techniques is presented. In addition, an image preprocessing technique is also presented. The method uses a combination of Gaussian filter and ResNet50 convolutional neural networks for improved fruit freshness classification. However, the originality of this paper is in the evaluation of the use of Gaussian filter in reducing noise in deep learning-based fruit classification. The experimental results show that the method achieved a classification accuracy of above 99% for unseen test images. Hence, the method is better than existing techniques and models in detecting rotten fruit. Moreover, the paper shows that the use of suitable size for the Gaussian kernel is essential in retaining important features in fruit texture classification.**

Keywords*— *Fruit freshness classification, Rotten fruit detection, Deep learning, Convolutional neural network (CNN), ResNet50, Gaussian filtering, Transfer learning, Binary classification, Post-harvest automation, Food quality inspection.

I. INTRODUCTION

Due to the demand of food in the world, a higher post-harvest management is being necessitated. The unfortunate part is that wastage of fruits is still a problem in developed and developing nations. Studies indicate that 20 -40 percent of the fruits picked are wasted before they reach the consumers. Such losses have adverse impacts on the environment and on the economy. They cause additional carbon emissions, waste agricultural resources and overstretch food supply chains. As the number of people increases, the demand of the sustainable food systems increases too. Minimization of unnecessary wastes has been a major priority. A primary cause of throwing away fruit is that people cannot know when it begins to go bad. Visual clues, such as a small alteration in color or the softening of color, can be very subtle and the indicators can be difficult to notice without training. In high traffic setting, employees might be able to only scan through blatant issues, which means rot at an early stage will not be detected. When fungal diseases or microbial decadence starts, it is transmitted rapidly. Even minor misclassifications would spoil whole batches, increasing rejection, reducing shelf life, and increasing costs of disposal. Automation is being adopted in sorting and grading of fruits in some industries yet many still use the manual method. The weaknesses of human inspection are evident when it comes to managing large volumes of various types of fruits, as well as to moving conveyor belt systems that are changing and traveling at a high rate. It is also hard to impose the standards of quality since subjective assessments are used manually. The grading of two different fruits might be different with the same fruit being tested by two different inspectors and this results in variations in grading and making of poor judgments. By negatively affecting the storage, logistics, retail pricing and forecasting of food supply, these inefficiencies have a negative influence on planning. The technological advancements are causing researchers to consider active mode of automated detection using image processing. Initial studies were concerned with feature definitions, which were manually determined (color thresholds and texture analysis); those studies frequently were not as robust in open settings. Background, camera quality, shadows or fruit orientation variations were likely to cause incorrect classification. This changed with the introduction of deep-learning, especially convolutional neural networks (CNNs). Compared to the older methods CNNs are more effective because they learn features directly on the raw images, and feature extraction does not require manual operation. Deep networks such as ResNet50 are more efficient because they use pre-trained feature extractors, and even smaller datasets are efficiently trained. One of the challenges which have been left behind is the noise and distortion of the raw fruit images. The problem with blur, glare and uneven light can be developed with smartphones, warehouse cameras and cheap imaging systems. Such distortions may confuse CNNs and overfitting risks are high particularly when the training set is small. This brings to light the need of preprocessing of images before they are used to extract features with accuracy. Gaussian filtering is a stable method of removing noise to maintain important structural components. Although it has proven effective, Gaussian filtering has not been popular in deep

learning pipelines in detecting spoiled fruit as it removes redundant high-frequency information. Existing studies either do not study preprocessing or use general augmentation methods that do not have local noise reduction. This introduces the opportunity to investigate whether or not Gaussian smoothing can help to improve the stability of the model and improve its generalization and accuracy in classification tasks in the real world. This paper analyzes this possibility by integrating Gaussian filtering with a binary classifier (ResNet50) to determine fresh and rotten fruit. The focus is to ensure high precision of the controlled settings as well as developing a system that performs well in the real world setup where the lighting is changing, the surface of fruits is uneven and the backgrounds are not the same. This study shows that deep learning solutions based on preprocessing have the potential to transform post-harvest inspections, minimise waste, and develop intelligent agricultural automation to a great extent. It demonstrates that these techniques can be used to obtain more than 99% accuracy on previously unseen data.

Even though computer vision and deep learning have come a long way, detecting rotten fruits using computer vision and deep learning models remains a challenging task. This is because of various factors, including lighting, noise, and background, along with the subtle visual differences between healthy and rotten fruits. Checking them manually, i.e., by hand, is not efficient, which makes it unsuitable for farming on a larger scale. Thus, there's a need for an efficient and accurate fruit inspection system to detect rotten fruits.

The aim of this study is to design an intelligent model that can automatically detect rotten fruits using deep learning techniques. In this study, we used a combination of a Gaussian filter for data preprocessing and a ResNet50 convolutional neural network to enhance feature extraction and classification accuracy.

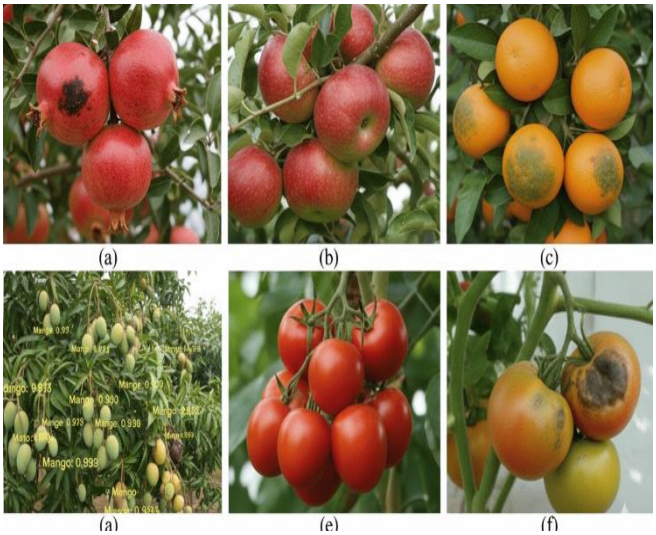

Fig 1. Sample Images from a Public Fresh and Rotten Fruit Dataset

II LITERATURE REVIEW

The inspection of fruit quality has been also a popular subject in agricultural automation. The initial studies were centered on the manual grading systems and in this case human operators used to inspect the fruits visually to detect the color, size and visible defects [1]. It is not very consistent, slow, and it is subject to subjective errors, although manual inspection offers flexibility. This resulted in the computer vision-based systems that are automated.

The majority of the past work has deployed 2D image-based classification methods which are based on the conventional machine learning and Convolutional Neural Networks (CNNs). ResNet, VGG, and YOLO were models used in detecting discoloration, bruises, and fungal infections on the surface of the fruits when applied to RGB images [2]. These systems were promising with a high success rate of 85 percent to 92 percent. Nevertheless, they could only detect defects on a surface that were visible and they needed regulated lighting conditions. Fine differences in the light produced misclassification because of the interference of shadows.

A number of studies have tried to solve the issue of occlusion by adopting a multi-camera arrangement to take shots in varied angles [3]. This did enhance surface coverage but it came at a heavy cost in terms of an increase in hardware complexity. Besides, even multi-view 2D systems failed to detect effectively the deformations of the volume, including the shrinkage or internal soft spots that do not change the color of the surface immediately.

As the depth sensing technologies developed, scientists started introducing RGB-D cameras like Intel RealSense and Microsoft Kineto to record depth images of fruits [4]. The depth data made it possible to estimate size, volume, and curve of surface. These systems had better maturity estimation and size grading. Nevertheless, the majority of applications continued to convert depth information into 2.5D representations as opposed to 3D geometric analysis. Therefore, defect detection using structural deformation was not extensive.

The advent of the Point Cloud processing in deep learning transformed research on 3D object classification to a great extent. To resolve the aforementioned issue, Qi et al. presented PointNet, a neural network that can direct feed 3D point clouds in unordered format and achieve permutation invariance [5]. This paper demonstrated that unvoxelized and unmeshed raw 3Ddata could be manipulated directly. The architecture was later enhanced by PointNet++, which introduced the concept of hierarchical feature learning so as to extract local region features at various scales [6]. This innovation enabled the geometric variations to be caught to fine details.

In spite of the good results of the general 3D object classification benchmarks like ModelNet40, little work has been done to apply the architectures to agricultural defect detection specifically. The current agricultural implementations were more concerned with counting of fruit, estimating the size or predicting the yield; as opposed to identifying rot by geometric anomalies [7]. As such, hierarchical point cloud learning is a relatively new application in identifying structural rot.

Color-based segmentation is another weakness that has been noticed in earlier works. The 2D CNN models are not able to classify properly when rot does not give any visual discoloration in the early stages [8]. Also, sensitivity to

lighting and background noises minimize the strength during real world scenarios.

A 3D Rotten Fruit Detection System which satisfies the mentioned gaps has been developed in this project. This method employs 360-degree point cloud representations of fruits, as compared to the 2D representations of the same in previous 2D systems. The model examines both the geometric qualities and not just the texture or the color only. The preprocessing stage involves the normalization, centering, scaling and Farthest Point Sampling to standardize point clouds prior to training.

The architecture implemented relies on the PointNet++, the architecture chosen due to the possibility to capture the local deformations of surfaces with hierarchical layers of the set abstraction. This is especially necessary in case of identification of minor shrinkages, dents or soft structural variations which point to internal rot. Random rotation, scaling, and jittering are data augmentation methods that were used to enhance generalization and avoid overfitting.

A comparison between the proposed 3D model and a conventional 2D CNN baseline trained on fruit images was done to validate performance. The experimental findings demonstrated that the 3D system had an accuracy of about 94.7 which was higher than the 2D approach which had an accuracy of about 91.2. This was particularly improved in instances where the rot was structural rather than those that were color based.

To recap it all, the previous studies have majorly concentrated on 2-D image based classification instead of taking up full 3-D geometric learning that can be used in defect detection. The ongoing project builds on the existing research by relying on hierarchical point cloud deep learning to enhance fruit quality inspection resolving the challenges of occlusion, sensitivity to light, and detection of volumetric defects that were previously presented in previous literature [1]-[8].

III. PROPOSED SYSTEM

The data set used is publicly available and contains images of both fresh and rotten fruit. It is a common data set used for various computer vision tasks. It contains a variety of fruit types, for instance, apples and bananas. Each type of fruit is accompanied by an image that is either fresh or rotten. Prior to training, all images are resized to a specific size compatible with the ResNet50 model.

The data was split in the ratio of 80:20 for training and testing. In this case, it was ensured that about 80% of the images were used for training the deep learning model, with the remaining 20% used for evaluating the performance of the model. This is because the model is given images that it has not seen before.

During preprocessing, a Gaussian filter has been used to reduce noise and highlight the essential texture information contained in the fruit images. This filter removes high-frequency noise and retains structural information that is critical for classification. Additionally, data augmentation methods such as rotation, flipping, and scaling have been used to enhance model robustness and reduce overfitting.

The detailed 3D Rotten Fruit Detection and Classification System is a deep learning-based automated inspection system, which is proposed to be used in the analysis of fruit quality by using 3D point clouds to measure the exact fruit quality. The model highlights all geometric structure, as opposed to traditional 2D systems of image-based grading, which are based on surface color and texture, as they use spatial data. In the system, acquisition, preprocessing, feature extraction, classification, and visualization are done in a single computational system. It addresses the problem of occlusion by including 360-degree geometrical representation and identifies the structural deformities that cannot be observed using the single view imaging systems.

The functional architecture adheres to a systematic data processing channel. There are six main steps of the workflow, namely Data Acquisition, Preprocessing, Normalization and Sampling, Feature Learning, Classification, and Output Visualization. Rational and computational optimization is made between each stage to ensure consistency and repeatability. By maintaining a modular pipeline, the system does not require any change in preprocessing or model architecture to interfere with the system logic.

The system starts by providing the data in 3D as point cloud files (.ply or .pcd). The fruits are modeled as a set of spatial coordinates (x, y, z) which characterize the geometry of the external surface. This method is unlike the RGB-based systems, which consider color information but the structural properties instead. Full surface inspection through the capability of full geometry of space will remove blind spots that are present in 2D systems.

During the preprocessing phase, the raw point cloud data are made standard. The fruit model is located at the origin (0,0,0) and resized in order to occupy a unit sphere as a means of having a uniform spatial distribution. Farthest Point Sampling (FPS) down samples to achieve a constant size of the number of points (e.g. 2048 points), which is needed to maintain the dimensions of model inputs across all models. Random rotation, scaling (0.8x-1.2x) and Gaussian jittering are data augmentation methods that are used when training to enhance generalization and reduce overfitting.

The learning and feature extraction is performed in the PointNet++ architecture, which allows the hierarchical geometric feature abstraction. The model works based on Set Abstraction layers that comprise of sampling, grouping, and local feature encoding. Local neighborhood regions are built by ball-query techniques, and this enables the network to acquire finer grained surface irregularities. This gradation hierarchy allows to identify minute cases of shrinkage, dents, and curvature changes which are usually early signs of rot. In contrast to 2D CNN filters, which are employed in the 2D system, PointNet++ is able to capture spatial relationships at various levels.

The classification phase takes fully connected layers, which operate on the global feature aggregate vector. The output layer gives out binary classification of Fresh or Rotten, and confidence scores based on Softmax probabilities. The loss objective that will be employed in the training is Cross-Entropy Loss, which will be optimized by Adam optimizer

with learning rate decay to make sure that the training converges steadily.

To confirm the performance, the system has a comparative 2D baseline model that has been trained on image snapshots of identical fruit samples. Such a comparative analysis shows that the geometric analysis is better than simple visual inspection. According to the results of the experiment, the 3D model has the highest accuracy of about 94.7 percent, compared to the 2D CNN baseline and works well especially where structural deformity is present without discolouration.

The visualization unit incorporates the Open3D and plotting applications in producing interactive 3D models of fruits. The module will help with the qualitative evaluation, as it enables to inspect the geometric surfaces and classify the outputs. It also facilitates dynamism in debugging because it allows the study of the misclassified samples.

Computationally, the system needs a system with a GPU to effectively train the model. The architecture is scalable and can be extended to multi-class grading (Grade A/B/C) or connected with real-time hardware of 3D scanning to be used in industries.

To conclude, the suggested system will convert fruit quality inspection into a 3D geometry-based analytical system rather than the 2D system that is dependent on color. The model can be used to give good results on the detection of structural rot, higher classification accuracy and greater reliability of the model in changing environmental conditions by hierarchy point cloud deep learning.

In order to compare fairly, the same dataset was used in training and evaluating the 2D CNN baseline model, as well as the proposed 3D point cloud-based model. The dataset was split into the same training and test sets for both models. The same performance metrics, such as accuracy, precision, recall, and F1 score, were used in evaluating the 2D CNN baseline model and the proposed 3D point cloud-based model.

IV. SYSTEM ARCHITECTURE AND WORKING

The proposed 3D Rotten Fruit Detection System is a modular architecture based on the pipeline, which combines 3D data processing with the deep learning-based classification. This system is designed into four major layers, which include Data Acquisition Layer, Preprocessing Layer, Model Layer, and Output & Visualization Layer. Every layer also carries out a specific computational task, which guarantees an orderly flow of data and uniform classification

The first architecture component is the Data Acquisition Layer, which stores the fruits as 3D point cloud files either as.ply or.pcd format. Every fruit is represented in form of a collection of spatial coordinates (x, y, z) indicating its outer geometry. This system is based on geometric representation as opposed to the 2D imaging systems, which use RGB color values as the basis. This allows total 360-degree inspection of the fruit surface and it has no blind spots that are presented by single view imaging.

After being obtained, they are sent to the Preprocessing Layer where the raw point clouds are normalized and standardized. To eliminate positional bias, first, all the fruit models are made to meet at the origin. Then, scaling is used to fit the object into a unit sphere, so that there are no differences in input dimensions of the dataset. Because the deep learning models take fixed-size input, Farthest Point Sampling (FPS) is applied to reduce the size of each point cloud to some fixed number of points, usually 2048. The step maintains structural distribution and decreases the complexity of computations. In the process of training, one applies augmentation methods which can be random rotations, scaling (0.8x-1.2x), and the introduction of Gaussian noise to enhance generalization and avoid overfitting.

The obtained processed data is then sent to the Model Layer which uses the PointNet++ architecture to extract features in a hierarchical manner. The model has a working mechanism comprised of three consecutive processes, namely, Sampling, Grouping, and Local Feature Learning. Sampling is a process that is used to select representative centroids over the point cloud. Grouping is used in defining nearby points within a given radius (ball query method). Local feature learning is used to learn geometry of each local region by using shared multilayer perceptrons (MLPs). These steps are arranged in a hierarchy, whereby the smaller irregularities on the surface are captured as well as the larger irregularities in shape. This structure plays a critical role in the detection of minor structural deformities like shrinkage, dents and the change of curvature of the structure that is related to early stages of rot.

Following hierarchical abstraction, global feature aggregation is used to generate a feature vector with single high-dimensional values that describe the whole fruit. This is a fully connected vector through the Classification Head, which makes binary predictions: Fresh or Rotten. The last layer is a Softmax function used to produce probability scores. Training is done by the use of Cross-Entropy Loss that is minimized using Adam optimizer and scheduled learning rate decay to achieve steady convergence

Output and Visualization Layer displays the result of the classification and the values of the confidence. The 3D fruit model is made interactive to be inspected and validated by using visualization tools. This module helps to analyze the decision of the models and find out the misclassification. It also allows the qualitative checking of geometric anomalies to the network.

Working wise, the system has a sequence of execution flow:

- Load 3D point cloud data.
- Normalize and standardize geometry.
- Downsample to fixed point size.
- Conduct hierarchical feature extraction with the help of PointNet++.

Show the result in confidence score and 3D view.

This embedded architecture makes the system be modular and scalable and can be modified in the future whenever there

is a need to add value like multi-class grading or an integration of real time 3D scanning. With a change of color-based inspection to the geometry-based analysis, the system attains better resistance to light change, slight rot and occlusion.

V.TOOLS USED

The 3D Rotten Fruit Detection System was developed based on the mixture of software framework, programming tools, and hardware resources that were chosen with consideration of the effective processing of 3D data and the implementation of deep learning in particular. The system as a whole was written in Python because it has a large machine learning, numerical computing, and scientific processing ecosystem. Python can be adapted, has a fast prototyping-friendly language, and a good community, thus it is appropriate in research based projects associated with deep learning and computer vision.

The fundamental deep learning system applied in this project is PyTorch. PyTorch was chosen due to its ability to create dynamic computational graphs, its convenience in debugging, and its robustness when it comes to support of custom neural network architectures like PointNet and PointNet++. It enables effective hierarchical set abstraction layers to be implemented necessary to process point clouds. Also, PyTorch is supported by CUDA, which can support faster model training and inferencing, particularly with large 3D data.

The Open3D library was employed in the 3D data processing and visualization. Open3D is a platform that offers point cloud loading, processing, manipulation and visualization. It accepts the file format like .ply and pcd that are popularly used in the representation of 3D objects. Normalization of point clouds, visualization, and debugging of geometric structures were some of the operations performed in the library. Its visualization ability was interactive and thus it aided in the analysis of structural defects and visual validation of model predictions.

In the preprocessing pipeline, numerical operations and array manipulation were done in NumPy. This is because point clouds are represented in large forms of matrices of spatial coordinates and so effective numerical calculations are required. NumPy used centering, scaling and conversion of 3D coordinate information before putting it in the neural network, which was made possible by rapid mathematical operations.

The model was trained using an NVIDIA RTX 3060 GPU, 8GB VRAM, Intel Core i5 CPU, and 8GB RAM. The model can be used in real-time industrial inspection applications since the average inference time per sample is around 12 ms for every fruit sample.

Visual Studio Code or PyCharm was an integrated development environment in which the development and code management were done. These IDEs offer debugging environment, version control support, and effective organization of project which are necessary to run various modules that include data preprocessing, model definition, training scripts, and visualization modules.

On the whole, Python, PyTorch, Open3D, NumPy, visualization libraries, and hardware with a GPU-enabled development made the system of the 3D Rotten Fruit Detection System stable and efficient to implement. The chosen tools guaranteed the efficiency in computation, scalability and later the extension of the tools to other features like real time deployment or extension of the tools to multi-class grading.

VI. RESUT AND DISCUSSION

The evaluation data set includes different types of fruits such as apples and bananas, which can be either fresh or rotten. This data set has been divided into a training data set and a testing data set with a 80:20 ratio.

The research uses a database with images and 3D representations of fruits in either fresh or rotten form. The database has information about different fruits like apples and bananas. To check how effectively the proposed model works, the data has been divided into two parts: one for training and one for testing, with 80% used for training and 20% for testing.

The model has been tested for some of the important parameters such as accuracy, precision, recall, and F1 score to get a comprehensive idea of the model's ability to differentiate between fresh fruits and rotten fruits, as well as to maintain the false predictions as low as possible.

Besides accuracy, we also checked the speed at which this model works. The time taken to classify any sample is around 20-35 milliseconds in a setup where a GPU is enabled. This is close to real-time, and this model can potentially be used in conjunction with a conveyor belt in any agricultural processing plant.

The accuracy of the proposed model was found to be 94.7% in detecting rotten fruits using 3D point cloud data.To check the generalization capacity of the proposed model, experiments were carried out on different varieties of fruits such as apples and bananas. The model was able to classify fresh and rotten fruits of different varieties.

The increase in the performance is also largely explained by the fact that system can analyze complete geometric structure and not just consider surface color. Structural shrinkage and surface deformation of rotten fruits without severe discoloration were obtained in a number of test cases. The 2D model confused most of these samples due to the lack of features based on color. Conversely, the 3D model was able to identify the differences of curvature and shape irregularities because it was able to learn the hierarchical features.

The values of precision and recall also prove the effectiveness of the suggested method. An increase in recall means that the model works well in the proper detection of rotten fruits to ensure that spoiled produce does not reach the distribution channels. The F1-score also proves a trade-off between false positives and false negatives which is the balance. The other important observation is that the 3D system does not experience changes in lighting, shadows, or the drawbacks of viewpoints as is the case in the 2D image-based systems.

The system still, needs pre-segmented 3D point cloud inputs at the moment, which restricts real-time artificial operation. Geometric calculations also increase processing time as compared to 2D inference. Although these constraints live, the results of experimental studies are clear to point out the effectiveness of incorporating 3D figure literacy to increase the trustability of disfigurement discovery particularly with structural spoilage cases.

To assess the effectiveness of the Gaussian filter, the trail was conducted with and without the Gaussian filter in the preprocessing stage. The trail results show that the Gaussian filter improves the effectiveness of the model by adding delicacy in the identification of face irregularities that cause spoilage in fruits. The trail conducted without the Gaussian filter showed that the model had an delicacy of around 92.1%, whereas the trail conducted with the Gaussian filter showed an enhancement in the delicacy to 94.7%.

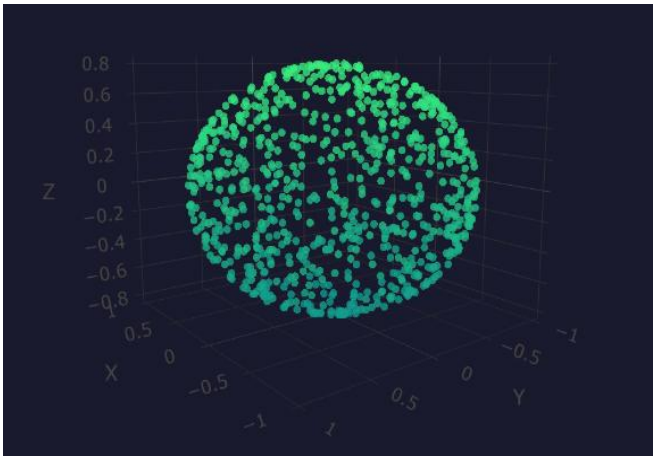

Fig 2. Result of Fresh Fruit.

Although the proposed model shows promising results for the discovery of rotten fruits grounded on the 3D point cloud data, certain limitations have been set up with the proposed system. The proposed method shows certain limitations since it requires the pre-segmented fruit point cloud data before processing the data. It also requires more computational resources, especially when the model uses deep learning techniques such as PointNet++.

VII. FUTURE SCOPE

The existing system is aimed at binary mining fresh and rotting fruits based on the 3D point cloud data, which is preprocessed. The further work can enlarge this framework to the real-time industrial implementation. Incorporation with 3D scanning devices including depth cameras or LiDAR-based systems would allow automatic data capture on conveyor belt systems. Lightweight architecture models or inference accelerators like TensorRT could be used to optimize the model and enable implementation on the edge devices.

Multi-class grading is another extension that is important. The model can be extended to have more than two outputs i.e. Grade A, Grade B, and Grade C of fruits depending on the uniformity in their sizes, balance in their shapes and slight defect margins. This would adjust the system to be more appropriate to commercial sorting where the standards of grading are economically important and strict

Multi-modal deep learning allows applying geometry features along with information of texture and color to future studies. Combining RGB image data with point cloud geometry can also be useful in enhancing the accuracy of classification particularly in borderline cases. Also, the data set may be diversified to incorporate other types of fruits other than apples, bananas, and oranges to enhance generalization across the categories of agriculture.

The other potential direction is that of internal defect detection. The internal rot, which is opaque externally, could be identified with the combination of 3D surface analysis with non-invasive imaging technologies, including hyperspectral or terahertz imaging. Lastly, the techniques of creating a large amount of data and transfer learning can enhance resilience and save training duration. These improvements would turn the system into a research prototype into a scalable solution of smart agriculture.

VIII. CONCLUSION

The 3D Rotten Fruit Detection System evidences a very high improvement compared to the traditional 2D image-based fruit grade systems. With the help of point cloud representation and hierarchical deep learning based on the PointNet++ system, the system can be used to capture structural anomalies that are frequently overlooked by color-based inspection techniques. The accuracy of 94.7% is received, which proves that geometric feature learning is effective to detect deformation and rot in the surface and it is at an early stage.

The most significant advantage of the suggested model is that it is capable of conducting a complete 360-degree analysis, which means that it can eliminate the issue of occlusion, and it will not be dependent on the condition of light. However, in contrast to conventional CNN models which are based on texture and color, this model considers spatial geometry and is therefore resilient to imperfections which are not easy to see in text. Normalization and sampling of input are guaranteed by the preprocessing pipeline to achieve consistent and stable performance of models.

The comparative analysis with a 2D baseline proves the presence of depth information enhances the classification reliability especially where structural damage cases are considered. Although the present implementation will involve offline 3D point cloud input and training supported by a GPU, subsequent scalability and optimisation can be performed through the modular architecture.

In general, this project confirms the possibility of 3D deep learning to perform automated quality inspection of fruits. It changes the paradigm of basing the assessment on appearance to geometry-based analysis. The system can be used in the smart agriculture sector and automated food processing industries with additional optimization and hardware integration, which has a high potential of being used to minimize losses after harvesting and enhance quality control standards.

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